

- A straight line of unit length drawn in the interval $(0,1)$ is divided into three intervals by choosing two points at random. The probability that the three line segments form a triangle is:

A) $\frac{1}{2}$ B) $\frac{1}{4}$ C) $\frac{2}{3}$ D) $\frac{1}{3}$
- If M is the median of a continuous random variable X and g is a decreasing function, then the median of $g(X)$ is:

A) $g\left(\frac{1}{M}\right)$ B) $g(M)$ C) $g^{-1}\left(\frac{1}{M}\right)$ D) $g^{-1}(M)$
- For discrete random variable X with probability mass function $p(x) = 2^{-x}$, $x = 1, 2, \dots$, the difference between the upper bound and actual value of $P(|X-2| > 2)$ is:

A) $\frac{1}{4}$ B) $\frac{3}{8}$ C) $\frac{5}{12}$ D) $\frac{7}{16}$
- Let X be an integer valued random variable with probability generating function $P(\cdot)$. Then $\sum_{n=0}^{\infty} s^n P(X \leq n) =$

A) $sP(s)$ B) $(2 - P(s))^{-1}$
C) $s(1 - s)^{-1}P(s)$ D) $(1 - s)^{-1}P(s)$
- Let X and Y be independent random variables assuming the values $1, 2, \dots, n$ with probability $\frac{1}{n}$, where $n > 1$. Then $P(X > Y)$ is equal to:

A) $\frac{1}{2}$ B) $\frac{1}{2n}$ C) $\frac{1}{2} - \frac{1}{2n}$ D) $\frac{1}{4} - \frac{1}{2n}$
- A continuous random variable X has raw moments $\mu'_r = r!$. Then the probability density function of X is:

A) $f(x) = e^{-x}, x > 0$ B) $f(x) = xe^{-x}, x > 0$
C) $f(x) = 1, 0 < x < 1$ D) $f(x) = e^{-|x|}, -\infty < x < \infty$
- Let $X_1, X_2, \dots, X_n$ be a random sample from the uniform distribution on the interval $(\theta, \theta + |\theta|)$. If $\theta \in (-\infty, 0)$ then the MLE of θ is:

A) $\frac{1}{2} \max_{1 \leq i \leq n} (X_i)$ B) $\max_{1 \leq i \leq n} (X_i)$ C) $\frac{1}{2} \min_{1 \leq i \leq n} (X_i)$ D) $\min_{1 \leq i \leq n} (X_i)$

8. Let $X_1, X_2, \dots, X_n$ be a random sample from a two point distribution with $P(X_1 = 1) = p$ and $P(X_1 = 0) = 1 - p$, $p \in (0,1)$. Let $T = \sum_{i=1}^n X_i$. The Baye's estimator of $p(1 - p)$ when the prior is the beta distribution with known parameter (α, β) , under squared loss function is:
- A) $\frac{T(\alpha+\beta-T)}{n(n-1)}$ B) $\frac{(T+\alpha+1)(n-T+\beta)}{(n+\alpha+\beta+2)(n+\alpha+\beta+1)}$
- C) $\frac{(T+\alpha+2)(n-T+\beta)}{(n+\alpha+\beta+1)(n+\alpha+\beta+2)}$ D) $\frac{(T+\alpha+1)(n-T+\alpha)}{(n+\alpha+\beta+1)(n+\alpha+\beta+2)}$
9. Let $X_1, X_2, \dots, X_n, n \geq 2$ be a random sample from $N(\mu, \sigma^2)$ with unknown μ and σ^2 . Then the power of one sample t test depends on:
- A) t distribution with n degrees of freedom
 B) non-central t distribution with n degrees of freedom
 C) t distribution with $n-1$ degrees of freedom
 D) non-central t distribution with $n-1$ degrees of freedom
10. Let $X_1, X_2, \dots, X_n$ be a random sample from the exponential distribution on the interval (θ, ∞) with scale parameter β . Let $\bar{X}$ and $X_{(1)}$ be the sample mean and first order statistic respectively. Suppose β is known. Then the likelihood ratio test of size α for testing $H_0: \theta \leq \theta_0$ against $H_1: \theta > \theta_0$, where θ_0 is a known constant has the form:
- A) $\phi(x) = \begin{cases} 1, & X_{(1)} < \theta_0 - n^{-1}\theta \log \alpha \\ 0, & X_{(1)} \geq \theta_0 - n^{-1}\theta \log \alpha \end{cases}$
- B) $\phi(x) = \begin{cases} 1, & X_{(1)} > \theta_0 - n^{-1}\theta \log \alpha \\ 0, & X_{(1)} \leq \theta_0 - n^{-1}\theta \log \alpha \end{cases}$
- C) $\phi(x) = \begin{cases} 1, & (\bar{X} - X_{(1)})/(\bar{X} - \theta_0) < \alpha^{n/(n-1)} \\ 0, & (\bar{X} - X_{(1)})/(\bar{X} - \theta_0) \geq \alpha^{n/(n-1)} \end{cases}$
- D) $\phi(x) = \begin{cases} 1, & (\bar{X} - X_{(1)})/(\bar{X} - \theta_0) > \alpha^{n/(n-1)} \\ 0, & (\bar{X} - X_{(1)})/(\bar{X} - \theta_0) \leq \alpha^{n/(n-1)} \end{cases}$
11. Let X be a single observation from the uniform distribution on $(\theta - \frac{1}{2}, \theta + \frac{1}{2})$, where $\theta \in R$ is unknown. Which among the following is/are true?
1. $\theta - X$ is a pivotal quantity.
 2. The distribution function of X is a non-decreasing function of θ .
- A) 1 only B) 2 only C) Both 1 & 2 D) Neither 1 nor 2

12. Let X and Y be discrete random variables, with jump points on $\{0,1,2,\dots\}$ and joint PMF $p(x,y) = \binom{y}{x} 2^{-y} e^{-\lambda} \frac{\lambda^y}{y!}$, $x = 0,1,\dots, y; y = 0,1,2,\dots, 0 < \lambda < \infty$. Which among the following statement(s) is/are true?
1. The marginal distribution of X is Poisson.
 2. The mean of the random variable Y is λ .
 3. The conditional distribution of X given Y is binomial.
- A) 1 only B) 1 & 2 only C) 2 & 3 only D) 1, 2 & 3
13. Let X_1 and X_2 be independent random variables having the standard normal distribution. Let $Y_1 = \sqrt{X_1^2 + X_2^2}$ and $Y_2 = \frac{X_1}{X_2}$. Which among the following statement(s) is/are true?
1. Y_1 has Weibull distribution.
 2. Y_2 has Cauchy distribution.
 3. Y_1 and Y_2 have a positive correlation.
- A) 2 only B) 1 & 2 only C) 2 & 3 only D) 1, 2 & 3
14. Let X be a random variable and let $g: R \rightarrow R$ be a Borel measurable function such that X and $Y = g(X)$ are independent. Then
- A) there exists a constant c such that $P[Y = c] = 1$
 - B) there exists a finite set B with cardinality greater than 1 such that $P[Y \in B] = 1$.
 - C) there exists an uncountable set B such that $P[Y \in B] = 1$.
 - D) none of the above statements are true.
15. Let (X, Y, Z) be a random vector with joint probability density function
- $$f(x,y,z) = \begin{cases} \frac{1 - \sin x \sin y \sin z}{\pi^3}, & 0 \leq x, y, z \leq 2\pi \\ 0, & \text{otherwise} \end{cases}$$
- Which among the following statement(s) is/are true?
1. The marginal distributions of X, Y and Z are uniform.
 2. X, Y and Z are pairwise independent but not independent.
 3. X, Y and Z are independent.
- A) 1 only B) 2 only C) 1 & 2 only D) 1 & 3 only
16. Let X be a random variable with $E(X^2) < \infty$ and let $Y = |X|$. Suppose that X has Lebesgue density symmetric about 0. Then
- A) $E(XY) = 1$.
 - B) X and Y are uncorrelated
 - C) X and Y have correlation equal to 1.
 - D) X and Y are independent.

17. Let X be a continuous random variable with probability density function $f(x) = 2x\exp(-x^2)$, $x \geq 0$, and zero otherwise. Then quantile of order p , $0 < p < 1$ of the distribution is:
- A) $(-\log(1-p))^{0.5}$ B) $(-\log(1-p))^2$
 C) $(-\log(p))^{0.5}$ D) $(-\log(p))^2$
18. Let Y be a discrete random variable which assumes nonnegative integer values. If the conditional distribution of X given $Y = y$ be the binomial distribution with parameter (y, p) , and Y has the Poisson distribution with mean θ , then the marginal distribution of X is:
- A) Poisson with mean $p\theta$ B) Poisson with mean $\frac{\theta}{p}$
 C) Poisson with mean $\frac{\theta}{1-p}$ D) Poisson with mean $(1-p)\theta$
19. The six words in the sentence I DO NOT LIKE THIS HOUSE are written on six identical slips and put in a box. A slip is selected at random. The average number of letters other than I in the word selected is:
- A) $\frac{8}{3}$ B) $\frac{20}{3}$ C) $\frac{10}{3}$ D) $\frac{13}{3}$
20. Let $X_1, X_2, \dots, X_n$ be independent and identically distributed random variables with $V(X_1) < \infty$. Then
- A) $\frac{1}{n(n+1)} \sum_{j=1}^n jX_j \xrightarrow{P} 0$ B) $\frac{1}{n(n+1)} \sum_{j=1}^n jX_j \xrightarrow{P} 1$
 C) $\frac{1}{n(n+1)} \sum_{j=1}^n jX_j \xrightarrow{P} E(X_1)$ D) $\frac{1}{n(n+1)} \sum_{j=1}^n jX_j \xrightarrow{P} E(X_1^2)$
21. Let $X_1, X_2, \dots, X_n$ be independent random variables satisfying $P[X_n = \pm n^\theta] = 0.5$ and $\theta > 0$ is a constant. If $T_n = \sum_{i=1}^n X_i$ then which of the following statement is true?
- A) When $\theta < 0.5$, $\frac{T_n}{n}$ converges to 0 a.s.
 B) When $\theta \geq 0.5$, $\frac{T_n}{n}$ converges to 0 a.s.
 C) When $\theta < 0.5$, $\frac{T_n}{n}$ converges to 0 in probability but does not converge to 0 a.s.
 D) When $\theta \geq 0.5$, $\frac{T_n}{n}$ converges to 0 in probability but does not converge to 0 a.s.
22. Let $X_1, X_2, \dots, X_n$ be random sample from $N(\mu, \sigma^2)$ with an unknown $\mu \in R$ and known $\sigma^2 > 0$. The UMVUE of μ^3 is:
- A) $\bar{X}^3$ B) $\bar{X}^3 - \frac{\sigma^2}{n}\bar{X}$ C) $\bar{X}^3 - \frac{2\sigma^2}{n}\bar{X}$ D) $\bar{X}^3 - \frac{3\sigma^2}{n}\bar{X}$

23. Let X be a random variable having gamma distribution with shape parameter α and scale parameter β , where α and β are unknown. Let $Y = \sigma \log X$. Which among the following statement(s) is/are true?
1. If $\sigma > 0$ is unknown, then the distribution of Y is in a location-scale family.
 2. If $\sigma > 0$ is known, then the distribution of Y is in an exponential family.
- A) 1 only B) 2 only C) Both 1 & 2 D) Neither 1 nor 2
24. Let $X_1, X_2, \dots, X_n$ be a random sample from the uniform distribution on the interval $[0,1]$ and let R be the sample range. The limiting distribution of $2n(1 - R)$ is:
- A) chi-square with 2 degrees of freedom.
 - B) chi-square with 4 degrees of freedom.
 - C) chi-square with n degrees of freedom.
 - D) chi-square with $2n$ degrees of freedom.
25. Let $X_1, X_2, \dots, X_n$ be a random sample from the uniform distribution on the interval $(0, \theta)$ with $\theta > 0$. Which among the following statement(s) is/are true?
1. The UMVUE of θ is a function of $\max(X_1, X_2, \dots, X_n)$.
 2. The variance of the UMVUE of θ is $\frac{\theta^2}{n+2}$.
 3. The variance of UMVUE of θ is equal to the Cramer-Rao lower bound.
- A) 1 only B) 1 & 2 only C) 1 & 3 only D) 2 & 3 only
26. Let $X_1, X_2, \dots, X_n$ be a random sample from the following discrete distribution:
 $P[X_1 = 1] = \frac{2(1-\theta)}{2-\theta}, P[X_1 = 2] = \frac{\theta}{2-\theta}$, where $\theta \in (0,1)$ is unknown. If $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$, then the moment estimator of θ is:
- A) $\bar{X} - \frac{1}{\bar{X}}$ B) $\frac{\bar{X}-1}{\bar{X}}$ C) $\frac{2(\bar{X}-1)}{\bar{X}}$ D) $\frac{\bar{X}-1}{2\bar{X}}$
27. The value of Karl-Pearson coefficient of correlation between two variables x and y is reported by a researcher as $r = -0.5$. Which of the following is correct?
- A) The x variable explains 25% of the variability in the y variable.
 - B) The x variable explains 50% of the variability in the y variable.
 - C) The x variable explains 5% of the variability in the y variable.
 - D) The x variable explains 95% of the variability in the y variable.
28. Let $X_1, X_2, \dots, X_n$ be a random sample from the distribution with probability density function $f_\theta(x) = \frac{1}{2}(1 + \theta x)$, $-1 \leq x \leq 1$, $|\theta| \leq 1$.
A consistent estimator of θ based on the above random sample is:
- A) $(n^{-1} \sum X_i)^{1/3}$ B) $(n^{-1} \sum X_i)^3$ C) $3n^{-1} \sum X_i$ D) $\frac{1}{3} n^{-1} \sum X_i$

29. Let $X_1, X_2, \dots, X_n$ be iid random variables having the probability density function $f_\theta(x) = \frac{2x}{\theta^2}, x \in [0, \theta]$. The MLE of the median of the distribution is:
 A) $\frac{\max X_i}{\sqrt{2}}$ B) $\sqrt{2} \max X_i$ C) $\frac{\min X_i}{\sqrt{2}}$ D) $\sqrt{2} \min X_i$
30. Let $X_1, X_2, \dots, X_{10}$ be iid Bernoulli random variables with probability of success equal to θ . The prior distribution of θ is given by $P[\theta = 0.5] = P[\theta = 0.9] = 0.5$. Assume that the realization of $(X_1, X_2, \dots, X_{10})$ is $(1, 1, 1, \dots, 1)$. The posterior probability for $\theta = 0.9$ is:
 A) $\frac{1}{1+(0.8)^{10}}$ B) $\frac{1}{1+(\frac{5}{9})^{10}}$ C) $\frac{1}{1+(\frac{5}{9})^{10} 5^5}$ D) $\frac{1}{1+(0.8)^5 (0.2)^5}$
31. A random sample of size 8 is taken from a distribution function $F(x)$ has the following values: 0, 1, 2, 2, 4, 6, 6, 7. The value of the empirical distribution function $F_8(x)$ when $2 \leq x < 4$ is:
 A) $\frac{1}{8}$ B) $\frac{1}{2}$ C) $\frac{5}{8}$ D) $\frac{7}{8}$
32. The following are random sample from two distribution functions:
 Sample 1: 1 2 3 5 7 9 11 18
 Sample 2: 4 6 8 10 12 13 14 15 19
 The value of the Mann-Whitney U statistic is:
 A) 36 B) 54 C) 72 D) 108
33. Consider the statistic D_n^+ of Kolmogorov-Smirnov one sample test for testing $H_0: F(x) = F_0(x)$ given $H_1: F(x) \geq F_0(x)$ based on a random sample of size n . The limiting distribution of $Y_n = 4n(D_n^+)^2$ is:
 A) chi-square with 1 degrees of freedom
 B) chi-square with 2 degrees of freedom
 C) chi-square with 4 degrees of freedom
 D) chi-square with 8 degrees of freedom
34. Suppose that $\{a_n\}_{n \geq 1}$ is a nonnegative sequence satisfying $\sum_{n=1}^{\infty} a_n = \infty$. Then $\sum_{n=1}^{\infty} \frac{a_n}{(a_1 + a_2 + \dots + a_n)^\alpha}$ is:
 A) converges when $\alpha > 1$ and diverges when $0 < \alpha \leq 1$.
 B) diverges when $\alpha > 1$ and converges when $0 < \alpha \leq 1$.
 C) converges for all values of α .
 D) diverges for all values of α .

35. Suppose that both $f(x)$ and $g(x)$ are monotone increasing continuous functions defined on $[0,1]$. Then

- A) $\int_0^1 f(x)dx \int_0^1 g(x)dx \geq \int_0^1 f(x)g(x)dx$
 B) $\int_0^1 f(x)dx \int_0^1 g(x)dx \leq \int_0^1 f(x)g(x)dx$
 C) $\int_0^1 f(x)dx \int_0^1 g(x)dx = \int_0^1 f(x)g(x)dx$
 D) none of the above relations is correct.

36. $\sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1} =$

- A) π B) $\frac{\pi}{2}$ C) $\frac{\pi}{3}$ D) $\frac{\pi}{4}$

37. Consider the sequence $\{x_n\}$ of real numbers defined by $x_1 = 1$ and $x_{n+1} = 1 + \frac{1}{1+x_n}$ for $n = 1, 2, \dots$. Then

- A) $\{x_n\}$ is a convergent sequence with $\lim x_n = 1$.
 B) $\{x_n\}$ is a convergent sequence with $\lim x_n = 2$.
 C) $\{x_n\}$ is a convergent sequence with $\lim x_n = \sqrt{2}$.
 D) $\{x_n\}$ is a divergent sequence.

38. Let $f: R \rightarrow R$ is Lebesgue measurable function such that $f(x+y) = f(x) + f(y)$ for all $x, y \in R$. Which among the following statement(s) is/are true?

1. f is continuous. 2. f has the form $f(x) = cx$.

- A) 1 only B) 2 only C) Both 1 & 2 D) neither 1 nor 2

39. Which among the following statement(s) is/are true?

1. If f is a measurable and integrable function then $|f|$ is also an integrable function.
 2. If f is a measurable and non-integrable function then $|f|$ is always a non-integrable function.

- A) 1 only B) 2 only C) Both 1 & 2 D) neither 1 nor 2

40. An engineering company advertises a job in three newspapers, A, B and C. It is known that these papers attract undergraduate engineering readerships in the proportions 2:3:1. The probabilities that an engineering undergraduate sees and replies to the job advertisement in these papers are 0.002, 0.001 and 0.005 respectively. Assume that the undergraduate sees only one job advertisement, and that the replies and readerships are independent. If the company receives two replies, then the probability that both applicants saw the job advertised in paper A is:

- A) $\frac{1}{3}$ B) $\frac{1}{4}$ C) $\frac{1}{9}$ D) $\frac{5}{12}$

41. Let $X_1, X_2, \dots, X_n$ be a random sample from $N(\mu, \sigma^2)$. Which of the following statement(s) is/are true for the likelihood ratio test for testing $H_0: \mu = \mu_0$ against $H_1: \mu \neq \mu_0$?
1. If σ^2 is known, the test is UMP.
 2. If σ^2 is unknown, the test is UMP.
- A) 1 only B) 2 only C) Both 1 & 2 D) neither 1 nor 2
42. In sampling from a distribution with mean 2 and variance 1, the asymptotic distribution of $\bar{X}^2$, where $\bar{X}$ is the sample mean is:
- A) normal distribution with mean 2
 - B) normal distribution with mean 4
 - C) chi-square distribution with 1 degrees of freedom
 - D) chi-square distribution with 2 degrees of freedom
43. Let $X_1, X_2, \dots, X_n$ be iid $N(\mu_1, \sigma_1^2)$ and $Y_1, Y_2, \dots, Y_n$ be iid $N(\mu_2, \sigma_2^2)$, and suppose that X 's and Y 's are independent. The variance of the sample correlation coefficient is:
- A) $\frac{1}{n-1}$ B) $\frac{1}{n-2}$ C) $\frac{\sigma_2^2}{\sigma_1^2} \frac{1}{n-1}$ D) $\frac{\sigma_2^2}{\sigma_1^2} \frac{1}{n-3}$
44. Let X be the total time (minutes) that a customer spends at a bank, and Y the time the customer spends waiting in line before reaching the teller. Assume that the customer gets into a line immediately after entering the bank. If the joint density of X and Y is $f(x, y) = \theta^2 e^{-\theta x}, 0 \leq y \leq x < \infty, \theta > 0$, and $f(x, y) = 0$, otherwise, then the mean service time is:
- A) θ B) 2θ C) $\frac{1}{\theta}$ D) $\frac{2}{\theta}$
45. Let X be the number of heads and Y the difference in absolute value between number of heads and number of tails in three independent tosses of a fair coin. The correlation coefficient between X and Y is:
- A) 0 B) 0.5 C) 0.6 D) 0.8
46. Consider the linear model $Y_{n \times 1} = X_{n \times p} \beta_{p \times 1} + \epsilon_{n \times 1}, \epsilon \sim N(0, \sigma^2 I)$. Let $l'Y$ be a linear function. Then the distribution of $\frac{(l'Y)^2}{\sigma^2 l'l}$ is:
- A) chi-square with 1 degrees freedom.
 - B) chi-square with $n-1$ degrees of freedom.
 - C) non-central chi-square with 1 degrees of freedom and non-centrality parameter $\frac{(l'X\beta)^2}{\sigma^2 l'l}$.
 - D) non-central chi-square with 1 degrees of freedom and non-centrality parameter $\frac{l'X\beta}{\sigma \sqrt{l'l}}$.

47. An engineer is interested in determining if the cotton weight percentage in a synthetic fiber affects the tensile strength and the engineer has run a completely randomized experiment with five levels of cotton weight percentage and five replicates. With usual notations, the model for the analysis is $y_{ij} = \mu + \tau_i + e_{ij}$, $i = 1, 2, \dots, 5$; $j = 1, 2, \dots, 5$. The values of the least square estimates of the parameters τ_1 , τ_3 and τ_5 , based on the following data are respectively.

Weight Percentage of Cotton	Observed Tensile Strength (lb/in ²)				
	1	2	3	4	5
15	7	7	15	11	9
20	12	17	12	18	18
25	14	18	18	19	19
30	19	25	22	19	23
35	7	10	11	15	11

- A) 0.36, -2.56 and -4.24 B) -5.24, 0.36 and 6.56
 C) 0.36, 6.56 and 0.36 D) -5.24, -2.56 and -4.24
48. Which of the following statement is **not** true for Kruskal-Wallis Test?
- A) Each group of observations in the data set are independent.
 B) Observations in each group come from populations with same distributional shape.
 C) The test cannot be used when there are ties in the data set.
 D) For reasonably large sample size the statistic of the test has chi-square distribution.
49. The following data is based on randomized block design, in which one observation is missing, and it is represented by x . The estimate of x that will have minimum contribution to error sum of squares is:

Treatments	Blocks			
	1	2	3	4
A	6.7	6.4	x	6.4
B	6.5	6.6	6.8	6.5
C	6.3	6.2	6.4	6.5

- A) 6.30 B) 6.50 C) 6.70 D) 6.85
50. Suppose that a 3^7 factorial design is conducted in 3^3 blocks by selecting 3 independent components of interactions to be confounded. The number of other components of interactions that are confounded automatically is equal to:
- A) 5 B) 8 C) 10 D) 16

51. Which of the following incomplete block designs is connected?

A)

Blocks	I	II	III	IV	V	VI	VII	VIII
Treatments	1	2	3	4	5	6	7	8
	3	4	5	6	7	8	1	2
	5	6	7	8	1	2	3	4

B)

Blocks	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII
Treatments	1	1	1	1	3	3	3	5	2	2	2	4
	3	5	8	8	5	7	8	7	4	4	4	6
	7	7	10	10	10	8	10	10	6	6	9	9

C)

Blocks	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII	XIII	XIV	XV
Treatments	1	1	1	1	2	2	2	2	3	3	3	4	4	5	5
	2	4	5	7	3	5	7	7	4	6	6	6	6	6	7

D) All the above are connected designs.

52. Let X be a random variable with probability density function proportional to $e^{-|x|^\alpha}$, where $0 < \alpha < 1$ and $-\infty < x < \infty$. Which of the following statement is true?

- A) MGF of X exists and hence all order moments exist.
- B) MGF of X does not exist, but moments of all order exist.
- C) MGF of X does not exist, but moments of order n exists only when $n > \alpha + 2$.
- D) MGF and moments of X do not exist.

53. For a bivariate data set (x_i, y_i) , $i = 1, 2, \dots, n$, the regression lines by the method of least squares are $8x - 10y + 66 = 0$ and $40x - 18y - 214 = 0$. Then which among the following statement(s) is/are true?

- 1. The value of the correlation coefficient is -0.6
- 2. Means of x values and y values are 13 and 17 respectively.
- 3. The ratio of standard deviation of y to standard deviation of x is $4/3$.

A) 1 only B) 2 & 3 only C) 1 & 2 only D) 1, 2 & 3

54. Which among the following statements are true for generalized power series distribution?

- 1. Negative binomial distribution is a special case of the generalized power series distribution.
- 2. Logarithmic series distribution is a special case of the generalized power series distribution
- 3. The equality of mean and variance is a necessary and sufficient condition for a power series distribution to be Poisson.

A) 1 & 2 only B) 2 & 3 only C) 1 & 3 only D) 1, 2 & 3

55. Let (v, b, r, k, λ) be the set of parameters of a balanced incomplete block design (BIBD). Which of the following (v, b, r, k, λ) **cannot** be the parameters of a BIBD?
- A) $(v, b, r, k, \lambda) = (6, 15, 10, 4, 6)$
 B) $(v, b, r, k, \lambda) = (5, 10, 6, 3, 3)$
 C) $(v, b, r, k, \lambda) = (17, 45, 8, 3, 1)$
 D) $(v, b, r, k, \lambda) = (9, 12, 4, 3, 1)$

56. If $\underline{X} \sim N_3(0, I)$ and $\underline{X}$ is partitioned as $\underline{X} = \begin{bmatrix} X_1 \\ X^{(2)} \end{bmatrix}$ where X_1 is of order 1×1 and $X^{(2)}$ is of order 2×1 , then which of the following statement/s is/are true?

1. $X_1 \sim N\left(0, \frac{1}{3}\right)$ 2. $X_1 \sim N(0, 1)$ 3. $X^{(2)} \sim N_2(0, I)$

- A) 1 only B) 2 only C) 1 & 3 only D) 2 & 3 only
57. Let $X_1, X_2, \dots, X_8$ be a random sample of size 8 from $N_5(\mu, \Sigma)$ and let $\bar{X} = \frac{1}{8} \sum_{i=1}^8 X_i$. Which of the following statements is false?
- A) $\bar{X}$ is an unbiased estimator of μ .
 B) $\bar{X}$ is the maximum likelihood estimator of μ .
 C) $\frac{1}{8} \sum_{i=1}^8 (X_i - \bar{X})(X_i - \bar{X})^T$ is an unbiased estimator of Σ
 D) $\frac{1}{8} \sum_{i=1}^8 (X_i - \bar{X})(X_i - \bar{X})^T$ is the maximum likelihood estimator of Σ

58. Let X be a 3×1 random vector. If the correlation matrix of its components is

$$\begin{bmatrix} 1 & \frac{1}{3} & \frac{1}{2} \\ \frac{1}{3} & 1 & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} & 1 \end{bmatrix}$$

then the partial correlation coefficient $\rho_{12.3}$ is:

- A) $\frac{1}{9}$ B) $\frac{1}{3}$ C) 0 D) 1
59. Let there be two multivariate normal populations. To test the equality of mean vectors of these populations, let us have two samples of sizes 8 and 12. If the value of Hotelling's T^2 statistic is obtained as 2.4, what will be the value of Mahalanobis' D^2 statistic?
- A) 207.36 B) 0.5 C) 1.52 D) 9
60. Let $\underline{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N_2(\mu, \Sigma)$ where $\mu = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 1 & 0.5 \\ 0.5 & 1 \end{bmatrix}$. Then $P[X_1 + X_2 \geq 7 + \sqrt{3}] = \dots$
- A) 0.8413 B) 0.8887 C) 0.1587 D) 0.1112

61. Let $\underline{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim N_2(\mu, \Sigma)$ where $\mu = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\Sigma = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$. Then the distribution of $X_1^2 + X_2^2 - 2(X_1 + X_2) + 2$ is:
- A) Bivariate normal B) Univariate normal
C) $\chi^2_{(1)}$ D) $\chi^2_{(2)}$
62. Let $\underline{X} \sim N_5(0, I)$ and A be a real symmetric matrix such that $A^2 = A$ and $\text{rank}(A) = 4$. Then the variance of $\underline{X}'A\underline{X}$ is
- A) 10 B) 5 C) 8 D) 4
63. If A is a matrix of order 3×3 with $|A| = 2$, then $|\text{adj}A| =$
- A) 2 B) 4 C) 8 D) 3
64. Which of the following statements is true?
(Here $\rho(A)$ denotes the rank of A , I_{10} denotes identity matrix of order 10 and O denotes null matrix.)
- A) $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ is idempotent
B) The determinant of an idempotent matrix is 1.
C) If A is an idempotent matrix of order 10, $\rho(A) + \rho(I_{10} - A) = 5$.
D) If $A^2 = O$ with $A \neq O$, then A is idempotent.
65. If A is an idempotent matrix with rank 6, then its trace is :
- A) 3 B) 6 C) 12 D) 5
66. Let A be a skew symmetric matrix of order 9. Which of the following statements is true?
- A) $\text{Rank}(A) = 9$ B) $\text{Rank}(A) = 8$
C) $\text{Rank}(A) < 9$ D) $\text{Rank}(A) < 8$
67. Identify the true statement/s:
- If λ is an eigenvalue of A , then $\frac{1}{\lambda}$ is an eigenvalue of A^{-1} .
 - If λ is an eigenvalue of A , then λ^2 is an eigenvalue of A^2 .
- A) 1 only B) 2 only C) Both 1 & 2 D) Neither 1 nor 2
68. Let W be a non – empty subset of a vector space $V(F)$. Let $a, b \in F$ and $\alpha, \beta \in W$. Identify the true statement/s:
- If W is a subspace of $V(F)$, then $a\alpha + b\beta \in W$.
 - If $a\alpha + b\beta \in W$, then W is a subspace of $V(F)$.
- A) 1 only B) 2 only C) Both 1 & 2 D) Neither 1 nor 2

69. Match the following:
- | | |
|------------------------|---|
| List I | List II |
| a. Idempotent matrix | 1. Eigen values are diagonal elements |
| b. Non-singular matrix | 2. Rank is the sum of its diagonal elements |
| c. Orthogonal matrix | 3. Inverse exists |
| d. Diagonal matrix | 4. Absolute value of its determinant is one |
- A) a-3, b-4, c-1, d-2 B) a-2, b-3, c-4, d-1
 C) a-1, b-2, c-3, d-4 D) a-4, b-3, c-1, d-2
70. Which of the following statements about two subspaces W_1 and W_2 of a vector space is **false**? (Here $\dim(W)$ denotes the dimension of the subspace W)
- A) $W_1 \cup W_2$ is a subspace.
 B) $W_1 \cap W_2$ is a subspace.
 C) $W_1 + W_2$ is a subspace.
 D) $\dim(W_1) + \dim(W_2) = \dim(W_1 \cap W_2) + \dim(W_1 + W_2)$
71. Brownian motion process is a :
- A) Stochastic process with discrete state space and continuous time space
 B) Stochastic process with continuous state space and continuous time space
 C) Stochastic process with discrete state space and discrete time space
 D) Stochastic process with continuous state space and discrete time space
72. Let $\{X(t), t \in T\}$ be a stochastic process such that the joint distribution of $[X(t_1), X(t_2), \dots, X(t_n)]$ is the same as that of $[X(t_1 + h), X(t_2 + h), \dots, X(t_n + h)]$ for all $h > 0$. Then $\{X(t), t \in T\}$ is called a:
- A) Markov process B) Weak stationary process
 C) Strong stationary Process D) Process with independent increments
73. The transition probability matrix of Markov chain $\begin{bmatrix} 0 & 1-2p & 2p \\ 2p & 0 & 1-2p \\ 1-2p & 2p & 0 \end{bmatrix}$ Then
- A) p is any real number
 B) p is any positive real number
 C) p is a real number less than 0.5
 D) None of the above
74. Which of the following statements are true in the case of a Poisson process?
1. The Poisson process possesses the Markov property.
 2. Sum of two independent Poisson processes is a Poisson process.
 3. Difference of two independent Poisson processes is not a Poisson process.
- A) 1 & 2 only B) 2 & 3 only C) 1, 2 & 3 D) None of these

75. The counter of a bank branch performs the transactions with a mean time of 2 minutes. The customers arrive at a mean rate of 20 customers/hour. If we assume that arrivals follow a Poisson process and that the service time is exponential, find the probability that counter is idle.
 A) $\frac{1}{3}$ B) $\frac{2}{3}$ C) $\frac{1}{10}$ D) None of these
76. Let $\bar{y}_1$ and $\bar{y}_2$ be the sample means based on a simple random sample of size n from a population of N units, under SRSWOR and SRSWR respectively.
 Let $S^2 = \frac{1}{N-1} \sum_{i=1}^N (Y_i - \bar{Y})^2$. Then $V(\bar{y}_2) - V(\bar{y}_1) =$
 A) $\frac{(n-1)S^2}{Nn}$ B) $\frac{(N-1)S^2}{Nn}$ C) $\frac{S^2}{Nn}$ D) $\frac{(n-1)S^2}{N-1}$
77. In stratified random sampling with a linear cost function of the form $C = C_0 + \sum_{h=1}^L C_h n_h$, where C = total cost, C_0 = overhead cost and C_h = cost per unit in the h^{th} stratum, the optimum stratum size n_h for the h^{th} stratum with usual notations which minimizes the variance of the stratified sample mean is:
 A) $n_h = n \frac{N_h S_h \sqrt{C_h}}{\sum_{h=1}^L N_h S_h \sqrt{C_h}}$ B) $n_h = n \frac{N_h S_h / \sqrt{C_h}}{\sum_{h=1}^L N_h S_h / \sqrt{C_h}}$
 C) $n_h = n \frac{N_h S_h \sqrt{C_h}}{\sum_{h=1}^L N_h S_h \sqrt{C_h}}$ D) $n_h = N_h S_h / C_h$
78. In cluster sampling, it is advisable to form clusters in such a way that
 A) Elements belonging to same cluster should be as homogeneous as possible and variation between clusters should be minimum
 B) Elements belonging to same cluster should be as heterogeneous as possible and variation between clusters should be maximum
 C) Elements belonging to same cluster should be as heterogeneous as possible and variation between clusters should be minimum
 D) None of the above
79. A researcher uses systematic sampling to select 30 units from a population consisting of 600 units serially numbered from 1 to 600. If he uses the random start as the 8th unit, which of the following unit will not be selected in the sample?
 A) 68 B) 348 C) 468 D) 578
80. A random sample of 20 units is taken from a population consisting of 500 units. Then finite population correction factor is:
 A) 0.96 B) 0.04 C) 24 D) 1